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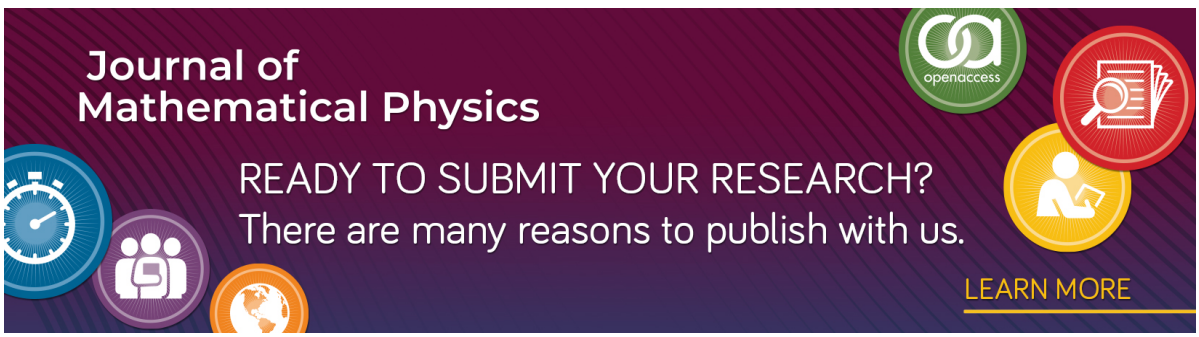
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ABSTRACT

We study analogs of Kuramoto and Cucker-Smale models for systems with infinite interacting particles. For such systems, we describe the way to derive the corresponding kinetic (mesoscopic) equation reflecting the statistical characteristics of the whole microscopic system in the sense of a proper scaling. This derivation has an algorithmic realization in terms of related hierarchical chains of correlation function equations. We compare the obtained mesoscopic equations for considered infinite interacting particle systems with kinetic equations derived from ordinary differential equation models of finite particle systems using the Bogoliubov–Born–Green–Kirkwood–Yvon hierarchy technique.

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I. INTRODUCTION

The collective behavior is ubiquitous in nature and society. Excellent examples of such behavior are synchronization of fireflies, neuron firing in the brain, flocking of birds, swarming of bacteria, and many others. Among others, synchronization and flocking seem to be the most interesting phenomena. Nowadays, there are many models attempting to understand synchronization and flocking from the mathematical point of view. The pioneering results of Winfree[29] and Kuramoto[23,24] launched a huge series of works about synchronization. The work[26] about synchronization of neurons or pacemaker cells deserves special attention. As for flocking, the fundamental work in which a mathematical model describing such an effect was presented belongs to Vicsek (see Ref. 28). Later, Cucker and Smale in Ref. 5 presented a simple and intuitive model describing the alignment of the velocity of each agent. The present work is devoted to the study of the analogs of the Kuramoto and Cucker-Smale (C-S) models for systems with infinite interacting particles.

The classical Kuramoto model describes dynamics of phases of $N \in \mathbb{N}$ oscillators, governed by the following ordinary differential equation (ODE) system:

$$\frac{d\theta_i}{dt} = \nu_i + \frac{\kappa}{N}\sum_{j=1}^{N} \sin(\theta_j - \theta_i), \quad \theta_i \in \mathbb{R}, \quad t > 0, \tag{1.1}$$

where θ_i and ν_i denote the phase and natural frequency of the ith oscillator. The nonnegative constant κ is called a coupling strength, measuring the intensity of interaction between oscillators.

On the other hand, the C-S model describes the velocity alignment phenomena of self-propelling agents. The C-S model is given by

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$$\begin{aligned}&\frac{dx_i}{dt} = v_i, \quad x_i, v_i \in \mathbb{R}^d, \quad t > 0, \quad 1 \le i \le N,\\ &\frac{dv_i}{dt} = \frac{\kappa}{N}\sum_{j=1}^{N} \phi(x_i - x_j)(v_j - v_i).\end{aligned} \tag{1.2}$$

Here, (x_i, v_i) denotes a position and velocity pair of the ith particle in d-dimensional Euclidean space, κ is a coupling strength as in the Kuramoto model, and $\phi : \mathbb{R}^d \to \mathbb{R}_+$ is called communication weight. It is a bounded and smooth function satisfying

$$\phi(x) = \phi(y) \quad \text{if} \quad |x| = |y|, \quad (\phi(x) - \phi(y))(|x| - |y|) \le 0, \quad \forall x, y \in \mathbb{R}^d.$$

When the number of particles is large enough, the dynamics of ensembles of particles can be effectively described by the corresponding kinetic equations. From the particle dynamics (1.1) and (1.2), it is possible to construct the partial differential equation for the one-particle density function f over the state space. One of the standard techniques to derive the governing kinetic equation is to use the Bogoliubov–Born–Green–Kirkwood–Yvon (BBGKY) hierarchy and molecular chaos assumption. Through the BBGKY hierarchy, we arrive at the kinetic equation for the Kuramoto model and the C-S model, respectively, as below. For the Kuramoto model, the resulting kinetic equation, sometimes called the Kuramoto-Sakaguchi equation, is given by

$$\begin{aligned}&\partial_t f + \partial_\theta(L[f]f) = 0, \quad (\theta, v) \in \mathbb{T} \times \mathbb{R}, \quad t > 0,\\ &L[f](\theta, v, t) := v - \kappa \int_{\mathbb{T}\times\mathbb{R}} \sin(\theta - \theta_*) f(\theta_*, v_*, t)\, d\theta_*\, dv_*,\\ &f(\theta, v, 0) = f_0(\theta, v),\end{aligned} \tag{1.3}$$

where $\mathbb{T}$ is the one-dimensional torus.

The kinetic description of the C-S model is a Vlasov-type equation which consists of a transport and a nonlocal flocking term,

$$\begin{aligned}&\partial_t f + v \cdot \nabla_x f + \nabla_v \cdot (F[f]f) = 0, \quad (x, v) \in \mathbb{R}^d \times \mathbb{R}^d, \quad t > 0,\\ &F[f](x, v, t) := \int_{\mathbb{R}^d\times\mathbb{R}^d} \phi(x - x_*)(v_* - v) f(x_*, v_*, t)\, dx_*\, dv_*,\\ &f(x, v, 0) = f_0(x, v).\end{aligned} \tag{1.4}$$

We present the details of deriving above mesoscopic equations, in particular, for the case of the C-S model, in the Appendix. Over the past years, the particle Kuramoto model (1.1), C-S model (1.2), and their corresponding kinetic equations (1.3) and (1.4) have been extensively studied from various perspectives. For example, a large-time behavior of each model is studied in Refs. 13 and 16 as well as the mean-field limit in Refs. 12 and 14. We refer to recent survey papers on the Kuramoto model[11] and the C-S model[4] for more detailed and numerous previous results.

In the present paper, we propose to study the statistical analogs of Kuramoto and C-S models as infinite interacting particle models which form a rich and powerful direction in modern stochastic and infinite dimensional analysis. One of the main ideas which are widely used in the study of complex systems lies in the mathematical approximation of a huge but finite real-world system by an infinite system realized in an infinite space.[7,9,20,21] This approach was successfully applied to the thermodynamic limit for models of statistical physics and appeared to be quite useful, e.g., for the ecological modeling in the infinite habitat to avoid boundary effects in a population evolution. Therefore, the phase space of the system should consist of countable sets from an underlying space. This space itself may have discrete or continuous nature that splits the world of complex systems into two big classes. Discrete models correspond to systems whose elements can occupy some prescribed countable set of positions, for example, vertices of the lattice $\mathbb{Z}^d$ or, more generally, of some graph embedded to $\mathbb{R}^d$. Continuous models, or models in the continuum, were studied less intensively and broadly. In the present paper, we essentially focus our attention on continuous counterparts of the Kuramoto and C-S models, whose elements (described, e.g., by the corresponding phases, positions, etc.) are points in Euclidean space $\mathbb{R}^d$. Having in mind that real elements have physical sizes, we will consider only the so-called locally finite subsets of the underlying space $\mathbb{R}^d$, which means that in any bounded region we assume to have a finite number of elements. Another restriction will be prohibition of multiple elements in the same position of the space.

A mathematical formalization of the abovementioned ideas regarding the phase space and time evolution of the considered systems can be described in the following way. As a phase space of the system, we use the space $\Gamma(\mathbb{R}^d)$ of locally finite configurations in the Euclidean space $\mathbb{R}^d$ (the dimension d has to be chosen properly according to the corresponding model). The mechanism of evolution of a system is supposed to be Markov, and it is initially given by a heuristic Markov generator on a suitable set of functions (observables) over $\Gamma(\mathbb{R}^d)$. It is very well known that the Markov evolution of a system gives rise to the evolution of observables (functional evolution) and evolution of distributions (state evolution). The first one is described by the Kolmogorov backward equation, and the second one is described by the Kolmogorov forward [also known as Fokker–Planck (FP)] equation on probability measures over $\Gamma(\mathbb{R}^d)$. It is worth noting that the question about the existence

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of the corresponding classical Markov processes on $\Gamma(\mathbb{R}^d)$ seems to be a challenging problem and up to our knowledge can be solved only in very special particular cases. As a result, a description of Markov dynamics in terms of random trajectories is *a priori* absent for most of the models under consideration. Hence, we use a concept of the statistical dynamics that substitutes the notion of a Markov stochastic process. This concept is based on the study of the FP equation only, which is apparently sufficient for many applications. The direct analysis of the FP equation is a very difficult mathematical problem in general. However, the FP equation can be reformulated as a hierarchical chain of equations for correlation functions of considered distributions. Such kinds of evolution equations are well known in the study of Hamiltonian dynamics for classical gases as BBGKY chains, but now they appear as a tool for construction and analysis of Markov dynamics. In the present paper, we do not study the problem of existence of statistical dynamics for the analogs of the Kuramoto and Cucker-Smale models for infinite interacting particle systems. These questions will be discussed in our forthcoming papers. Instead, we propose to study such models using the notion of the Vlasov scaling, provided that evolution exists on the microscopic level. Note that the traditional approach to the dynamical theory of large systems such as ecological communities, economical societies, cells, or virus populations is based on one of the three different levels of the system's description (at microscopic, mesoscopic, or macroscopic scales) with its own logic and own history. The microscopic theory deals directly with the elements in terms of their phase-space variables and describes the evolution of such a system as a stochastic Markov dynamics, whereas the two other levels involve a scaled representation, which reduces the complexity of the underlying microscopic system. The intermediate (mesoscopic) theory is constructed via suitable scaling limits and provides a simplified description of the collective behavior of individual particles. The highest description, macroscopic, treats the system as a whole, and it is based upon temperature and space/time-dependent macroscopic parameters such as density. Of course, the microscopic approach provides the best possible description, but it is sometimes related with very difficult mathematical problems. Usually, the mesoscopic description (which is the key point in the present paper) is evidently vague for the original particle model as it ignores the original structure of the system. However, the mesoscopic description has a great advantage since it can be mathematically analyzed in a much simpler way. The Vlasov equation is a famous example of a kinetic equation which describes the dynamical behavior of a system on the mesoscopic level. For a general overview in this context, we refer to Ref. 10.

To summarize, the main goal of this paper is twofold. First of all, we introduce the analogs of the Cucker-Smale and Kuramoto models for infinite particle systems by deriving the corresponding (FP) equations. Second, using the Vlasov scaling approach, we derive the mesoscopic equations governing those collective dynamics models. As intuitively expected, we show that the kinetic equations for the analogs of the Cucker-Smale and Kuramoto models for infinite particle systems coincide with those derived from the usual BBGKY hierarchy for finite systems.

II. PRELIMINARIES

In this section, we provide a general setup for continuous infinite particle systems. Let $\mathcal{B}(\mathbb{R}^d)$ be the Borel σ-algebra and $\mathcal{B}_b(\mathbb{R}^d)$ be the class of bounded Borel sets.

The natural phase space for the stochastic models considered in the present paper is the space of all locally finite subsets of $\mathbb{R}^d$. It is defined by

$$\Gamma = \Gamma(\mathbb{R}^d) := \Big\{\gamma \subset \mathbb{R}^d \ : \ |\gamma \cap K| < \infty \text{ for any compact set } K \subset \mathbb{R}^d\Big\}.$$

This space is naturally embedded into the set of all Radon measures $\mathbb{M}(\mathbb{R}^d)$ via

$$\Gamma(\mathbb{R}^d) \hookrightarrow \mathbb{M}(\mathbb{R}^d), \ \gamma \mapsto \sum_{x\in\gamma} \delta_x.$$

Hence, $\Gamma(\mathbb{R}^d)$ can be equipped with the vague topology, i.e., the weakest topology for which all mappings

$$\Gamma \ni \gamma \to \sum_{x\in\gamma} f(x) \in \mathbb{R}$$

are continuous for any continuous function f on $\mathbb{R}^d$ with compact support. The topological space $\Gamma(\Lambda)$ for any $\Lambda \in \mathcal{B}_b(\mathbb{R}^d)$ can be defined in a similar way.

The Borel σ-algebra $\mathcal{B}(\Gamma)$ on Γ is defined as the smallest σ-algebra such that all mappings

$$N_\Lambda : \Gamma \to \mathbb{N}_0, \quad N_\Lambda(\gamma) = |\gamma \cap \Lambda|$$

are measurable. This σ-algebra can be generated by the sets

$$\Gamma(\Lambda, n) := \big\{\gamma \in \Gamma | N_\Lambda(\gamma) = n\big\}, \qquad \Lambda \in \mathcal{B}_b(\mathbb{R}^d), \quad n \in \mathbb{N}_0. \tag{2.1}$$

Clearly, for any $\Lambda \in \mathcal{B}_b(\mathbb{R}^d)$,

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$$\Gamma = \bigsqcup_{n\in\mathbb{N}_0} \Gamma(\Lambda, n).$$

For the technical purposes related to the approach which we use in our paper, we will also be interested in the space of finite configurations whose natural topology is different from the vague one considered on Γ. The construction of the space of finite configuration is given as follows. As the initial step, we define the space of n-point configurations,

$$\Gamma_0^{(n)} = \Gamma_{0,\mathbb{R}^d}^{(n)} := \left\{\eta \subset \mathbb{R}^d \,\middle|\, |\eta| = n\right\}, \quad n \in \mathbb{N}_0 := \mathbb{N} \cup \{0\},$$

where $|A|$ denotes the number of elements in A and $\Gamma_{0,\mathbb{R}^d}^{(0)} := \{\emptyset\}$. Note that the set $\Gamma_0^{(n)}$ is equivalent to the symmetrization of

$$\widetilde{(\mathbb{R}^d)^n} := \left\{(x_1, \ldots, x_n) \in (\mathbb{R}^d)^n \,|\, x_k \neq x_l, \quad \text{if } k \neq l\right\},$$

i.e., $\widetilde{(\mathbb{R}^d)^n}/S_n$, where S_n is the permutation group on the set $\{1, 2, \ldots, n\}$. Therefore, $\Gamma_{0,\mathbb{R}^d}^{(n)}$ naturally inherits the topology $O(\Gamma_0^{(n)})$ of the space $\widetilde{(\mathbb{R}^d)^n}/S_n$ and the Borel σ-algebra $\mathcal{B}(\Gamma_0^{(n)})$ is the one induced by that of $(\mathbb{R}^d)^n$. Now, we collect the sets of n-point configurations and define the set of all finite configurations,

$$\Gamma_0 := \bigsqcup_{n\in\mathbb{N}_0} \Gamma_0^{(n)},$$

which is equipped with the topology of the disjoint union. We denote the corresponding Borel σ-algebra to this topology by $\mathcal{B}(\Gamma_0)$. The space of n-point configurations in $Y \in \mathcal{B}(\mathbb{R}^d)$, denoted by $\Gamma_{0,Y}^{(n)}$, can be defined analogously to $\Gamma_{0,\mathbb{R}^d}^{(n)}$. For the space of finite configurations in $Y \in \mathcal{B}(\mathbb{R}^d)$, we will use the symbol $\Gamma_{0,Y}$. It is worth noting that Γ_0 is a subset of Γ in terms of sets. However, topological spaces Γ_0 and Γ are not comparable.

Next, we describe the classes of functions and measures on Γ_0 and Γ which will be used in the sequel. A set $M \in \mathcal{B}(\Gamma_0)$ is called bounded if there exists $\Lambda \in \mathcal{B}_b(\mathbb{R}^d)$ and $N \in \mathbb{N}$ such that $M \subset \sqcup_{n=1}^N \Gamma_{0,\Lambda}^{(n)}$. The class $B_{bs}(\Gamma_0)$ stands for the set of all bounded measurable functions on Γ_0 which have bounded support, i.e., $G \in B_{bs}(\Gamma_0)$ if $G|_{\Gamma_0\setminus M} = 0$ for some bounded $M \in \mathcal{B}(\Gamma_0)$. The class $\mathcal{F}L^0(\Gamma)$ denotes the set of cylindrical functions on Γ, i.e., the set of all measurable functions F on $(\Gamma, \mathcal{B}(\Gamma))$, which are measurable with respect to $\mathcal{B}(\Gamma(\Lambda))$ for some $\Lambda \in \mathcal{B}_b(\mathbb{R}^d)$. Any $F \in \mathcal{F}L^0(\Gamma)$ is determined by a set $\Lambda \in \mathcal{B}_b(\mathbb{R}^d)$ such that $F(\gamma) = F(\gamma_\Lambda)$ for all $\gamma \in \Gamma$. A more detailed description of these classes of function can be found in Ref. 17.

Let σ be a nonatomic Radon measure on $\left(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d)\right)$, e.g., the Lebesgue measure. The analog of the Lebesgue measure on $(\Gamma_0, \mathcal{B}(\Gamma_0))$ is the Lebesgue-Poisson measure. It is defined on $(\Gamma_0, \mathcal{B}(\Gamma_0))$ by

$$\lambda_z := \delta_\emptyset + \sum_{n=1}^{\infty} \frac{z^n}{n!} m^{(n)}, \quad z > 0,$$

where $m^{(n)}$ is the restriction of the product measure σ^n considered on $(\mathbb{R}^d)^n$ to $(\Gamma_0^{(n)}, \mathcal{B}(\Gamma_0^{(n)}))$. Throughout the paper, we take the parameter z to be 1 and σ to be the Lebesgue measure using the notation $\lambda := \lambda_1$ for this case.

By $\mathbb{M}^1(\Gamma)$, we denote the class of all probability measures on $\left(\Gamma, \mathcal{B}(\Gamma)\right)$. Let us consider a collection of random variables $N_\Lambda(.)$, $\Lambda \in \mathcal{B}_b(\mathbb{R}^d)$ on probability space $\left(\Gamma, \mathcal{B}(\Gamma)\right)$ with $\mu \in \mathbb{M}^1(\Gamma)$. They describe random numbers of elements of configurations inside of bounded regions. The natural assumption for particle distributions is that these random variables should have finite moments. Thus, we will be interested in the class $\mathbb{M}^1_{\text{fm}}(\Gamma)$ of all measures from $\mathbb{M}^1(\Gamma)$ such that

$$\int_\Gamma |\gamma_\Lambda|^n \, d\mu(\gamma) < \infty, \qquad \Lambda \in \mathcal{B}_b(\mathbb{R}^d), \quad n \in \mathbb{N}. \tag{2.2}$$

An excellent example of such a measure (basic measure on configuration space) is the *Poisson measure* π_σ on Γ with intensity measure σ. It is defined on $(\Gamma, \mathcal{B}(\Gamma))$ by

$$\pi_\sigma\left(\Gamma_{0,\Lambda}^{(n)}\right) = \frac{\left(\sigma(\Lambda)\right)^n}{n!} \exp\left\{-\sigma(\Lambda)\right\}, \qquad \Lambda \in \mathcal{B}_b(\mathbb{R}^d), \quad n \in \mathbb{N}_0. \tag{2.3}$$

In the sequel, we use notation π instead of π_σ if σ is the Lebesgue measure on $\left(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d)\right)$. The Poisson measure plays the role of a chaotic, i.e., free state, where all elements of configuration are not correlated.

Now, we introduce the following mapping between $B_{bs}(\Gamma_0)$ and $\mathcal{F}L^0(\Gamma)$, which turns out to be crucial studying functions and measures on Γ_0 and Γ. For any function $G \in B_{bs}(\Gamma_0)$, we define the K-transform of G as

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$$KG(\gamma) := \sum_{\eta \Subset \gamma} G(\eta), \quad \gamma \in \Gamma. \tag{2.4}$$

Here, the notation $\eta \Subset \gamma$ means that η is a finite subconfiguration of γ. Note that the mapping K is well defined, i.e., $KG \in \mathcal{F}L^0(\Gamma)$ for $G \in B_{bs}(\Gamma_0)$. It is linear, positivity preserving, and invertible with inverse function K^{-1} which is given by the following equation:

$$K^{-1}F(\eta) := \sum_{\xi \subset \eta} (-1)^{|\eta \setminus \xi|} F(\xi), \quad F \in \mathcal{F}L^0(\Gamma), \quad \eta \in \Gamma_0. \tag{2.5}$$

Moreover, the K-transform can be extended to a bijective operator K: $L^1(\Gamma_0, d\lambda) \to L^1(\Gamma, d\pi)$. For more details about the properties of K, see Refs. 1 and 17.

A measure $\mu \in \mathbb{M}^1(\Gamma)$ is called locally absolutely continuous with respect to the Poisson measure π if for any $\Lambda \in \mathcal{B}_b(\mathbb{R}^d)$ the projection of μ onto $\Gamma(\Lambda)$ is absolutely continuous with respect to the projection of π onto $\Gamma(\Lambda)$. More precisely, if we consider the projection mapping $p_\Lambda : \Gamma \to \Gamma(\Lambda), p_\Lambda(\gamma) := \gamma_\Lambda$, then $\mu^\Lambda := \mu \circ p_\Lambda^{-1}$ is absolutely continuous with respect to $\pi_\Lambda := \pi \circ p_\Lambda^{-1}$.

According to, e.g., Ref. 17, for any $\mu \in \mathbb{M}^1_{\mathrm{fm}}(\Gamma)$ which is locally absolutely continuous with respect to π, there exists a *correlation function* $k_\mu : \Gamma_0 \to \mathbb{R}_+$ which is defined as follows. For all $G \in B_{bs}(\Gamma_0)$,

$$\int_\Gamma (KG)(\gamma) d\mu(\gamma) = \int_{\Gamma_0} G(\eta) k_\mu(\eta) d\lambda(\eta). \tag{2.6}$$

As was shown in Ref. 17, the right hand side of (2.4) can be extended to all G such that the left hand side of (2.6) is finite. In this case, (2.4) holds for μ-a.a. $\gamma \in \Gamma$ and (2.6) is fulfilled.

The notion of the correlation function plays a crucial role in the study of the FP equation for infinite particle systems. It allows us to study the FP equation by means of the infinite system of evolution equations considered in the functional spaces over $(\mathbb{R}^d)^n$, $n \in \mathbb{N}$. The latter fact becomes more clear if we consider the following remark.

Remark 2.1. An important feature of a correlation function $k : \Gamma_0 \to \mathbb{R}$ is that it can be written as a sequence of functions,

$$k^{(n)} : (\mathbb{R}^d)^n \to \mathbb{R}_+, \; k^{(n)}(x_1, \dots, x_n) = \begin{cases} k(\{x_1, \dots, x_n\}), & \text{if}(x_1, \dots, x_n) \in \widetilde{(\mathbb{R}^d)^n}, \\ 0, & \text{otherwise}, \end{cases}$$

which are called n-point correlation functions.

The main technical observation that is used in the analysis of the analogs of the Kuramoto and Cucker-Smale models for systems with infinite interacting particles is that of Minlos.

Lemma 2.1 (Minlos lemma[25]). *Let $n \in \mathbb{N}$, $n \geq 2$, and $z > 0$ be given. Then, for all measurable functions $G : \Gamma_0 \to \mathbb{R}$ and $H : \Gamma_0 \times \cdots \times \Gamma_0 \to \mathbb{R}$, we have*

$$\int_{\Gamma_0} \cdots \int_{\Gamma_0} G(\eta_1 \cup \cdots \cup \eta_n) H(\eta_1, \cdots \eta_n) d\lambda_z(\eta_1) \cdots d\lambda_z(\eta_n)$$
$$= \int_{\Gamma_0} G(\eta) \sum_{(\eta_1, \dots, \eta_n) \in \mathcal{P}_n(\eta)} H(\eta_1, \dots, \eta_n) \, d\lambda_z(\eta),$$

provided both sides of the equality make sense. Here, $\mathcal{P}_n(\eta)$ is the set of all ordered partitions of η into n subconfigurations.

III. EVOLUTION OF DISTRIBUTIONS AND KINETIC EQUATIONS

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. Consider a Markov process X_t^γ with values in configuration space Γ and a starting configuration $\gamma \in \Gamma$. Suppose that the initial configuration is distributed by $\mu_0 \in \mathbb{M}^1(\Gamma)$. Then, the distribution of X_t^γ given by $\mu_t := \mathbb{P}((X_t^\gamma)^{-1})$ satisfies the following Kolmogorov forward [also known as Fokker–Planck (FP)] equation:

$$\frac{\partial}{\partial t} \langle F, \mu_t \rangle = \langle LF, \mu_t \rangle, \quad \mu_t|_{t=0} = \mu_0, \tag{3.1}$$

where $F : \Gamma \to \mathbb{R}$ belongs to the domain of the corresponding Markov generator L and the duality is given by

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$$\langle F, \mu \rangle := \int_{\Gamma} F d\mu.$$

This duality relates the FP equation to the Kolmogorov backward equation which describes the evolution of observables (functional evolution),

$$\frac{\partial}{\partial t} F_t = LF_t, \quad F_t|_{t=0} = F. \tag{3.2}$$

From the point of view of applications, dealing with huge systems of interacting particles, it is not necessary and realistic to know all positions of the process at any moment of time (in other words, to know all trajectories). It is rather important to know all statistical characteristics of such a process, e.g., density, correlations of different orders, etc. Hence, to study the Markov evolution of infinite interacting particle systems, we use a concept of statistical dynamics that substitutes a Markov stochastic process (by means of trajectories) by the evolution of initial distribution [as a solution to the FP equation]. Let us assume that for each moment of time t, the distribution μ_t has the correlation function $k_t := k_{\mu_t}$ given by (2.6). Then, the evolution μ_t can be described by the evolution of k_t. Similar to (3.1), the correlation function k_t will satisfy the following equation:

$$\frac{\partial}{\partial t} \langle\langle G, k_t \rangle\rangle = \langle\langle \hat{L} G, k_t \rangle\rangle, \quad k_t|_{t=0} = k_0, \quad G \in D(\hat{L}), \tag{3.3}$$

where formally $\hat{L} := K^{-1} L K$ (*symbol* or *K-image* of L) and the duality is given by

$$\langle\langle G, k \rangle\rangle = \int_{\Gamma_0} G(\eta) k(\eta) \lambda(d\eta). \tag{3.4}$$

Note that $D(\hat{L})$ in (3.3) has to be chosen consistently with the domain of L and (2.6). It is worth pointing out that if only the initial k_0 is a correlation function of some probability measure $\mu_0 \in \mathbb{M}^1(\Gamma)$, then the solution k_t to (3.3) would not necessarily be the correlation function of some probability measure on Γ. The sufficient conditions which ensure that the latter fact holds, i.e., there exists a family $\{\mu_t\}_{t \geq 0}$ corresponding to the evolution of (3.3), can be found in, e.g., Ref. 17. We may also write Eq. (3.3) in its strong form,

$$\frac{\partial}{\partial t} k_t(\eta) = L^{\triangle} k_t(\eta), \quad k_t|_{t=0} = k_0, \tag{3.5}$$

where $L^{\triangle} = \hat{L}^*$ is adjoint to $\hat{L}$ with respect to (3.4) if it exists. For some technical purposes, one may also be interested in the predual evolution given by

$$\frac{\partial}{\partial t} G_t = \hat{L} G_t, \quad G_t|_{t=0} = G_0. \tag{3.6}$$

The considered functions are also called quasiobservables, since they are of purely mathematical interest, but have a direct correspondence to observables $F : \Gamma \to \mathbb{R}$.

In the present paper, we do not study the question about the existence of a solution to (3.1) for the infinite particle statistical counterparts of the Kuramoto and Cucker-Smale models. We suppose that the solution $\{\mu_t\}_{t \geq 0}$ to (3.1) exists. Moreover, we assume that the corresponding evolution of correlation functions $\{k_t\}_{t \geq 0}$ satisfying either (3.3) or (3.5) exists too. Our aim is to find the mesoscopic description for the statistical analogs of the Kuramoto and Cucker-Smale models for infinite systems of particles. For this purpose, we now outline the general approach to the mesoscopic (Vlasov) scaling. The Vlasov equation is a famous example of a kinetic equation which describes the dynamical behavior of a many-body system; see, e.g., Ref. 27. In physics, it characterizes the Hamiltonian motion of an infinite particle system influenced by weak long-range forces in the mean field scaling limit. The convergence in the Vlasov scaling limit was shown by Braun and Hepp[3] (for the Hamiltonian dynamics) and by Dobrushin[6] (for more general deterministic dynamical systems). The limiting Vlasov-type equations for particle densities are usually considered in classes of integrable functions (or finite measures in the weak form). This corresponds, actually, to the situation of finite volume systems or systems with zero mean density in an infinite volume. A detailed analysis of Vlasov-type equations for integrable functions was presented in the recent paper by Kozlov.[22] Motivated by the study of Vlasov scaling for some classes of stochastic evolutions in the continuum for which the use of the abovementioned approaches breaks down (even in the finite volumes), we use the general approach to study the Vlasov-type dynamics proposed in Ref. 10. It is based on a proper scaling of the hierarchical equations for the evolution of correlation functions and can be interpreted in terms of the rescaled Markov generators. The main difficulty to carry out the approaches proposed by Braun, Hepp, and Dobrushin for general stochastic evolutions in continuum models is the absence of the suitable descriptions in terms of stochastic evolution equations. Contrary to the classical derivation of the Vlasov-type kinetic equations from the Hamiltonian dynamics, we do not prove the law of large numbers for the corresponding processes. We do not even need to show the existence of the corresponding microscopic rescaled processes. Our main idea is to study the evolution of states (distributions) of the system in terms of the corresponding chain of hierarchical equations. As pointed out by Spohn,[27] the correct Vlasov limit can be easily guessed from the BBGKY

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hierarchy for the Hamiltonian system. Such a heuristic derivation does not assume the integrability condition for the density, but until now, it could not be made rigorously due to the lack of detailed information about the properties of solutions to the BBGKY hierarchy. We would like to stress that different classes of initial data are not only mathematical tools for the rigorous study of the problem. They describe different physical situations in related microscopic models. The zero average density systems were considered in Ref. 2 by means of heuristic limit transition in the corresponding hierarchical equations for correlation functions. The framework we are usually working in is nonzero average density which is related to the case of bounded correlation functions. Our approach is based on Spohn's observation applied in a new dynamical framework. More precisely, we already know that many stochastic evolutions in continuum admit effective descriptions in terms of hierarchical equations for correlation functions which generalize the BBGKY hierarchy from Hamiltonian to Markov setting. Moreover, these hierarchical equations are often the only available technical tools for the construction of corresponding dynamics in several models.[8,18,19]

For the reader's convenience, we outline below the general scheme to derive the kinetic equation for the infinite interacting particle system described by (3.1). The first step in the Vlasov scaling is to scale the intensities of the interaction of the system; usually, one dumps the potentials appearing in L by a factor $\varepsilon > 0$. In general, the exact scaling of L is carried out for each model separately. The second step is to increase the density of the system, i.e., we consider initial conditions $k_{0,\varepsilon} = (k_{0,\varepsilon}^{(n)})_{n\in\mathbb{N}_0}$, which satisfy

$$\varepsilon^n k_{0,\varepsilon}^{(n)} \to r_0^{(n)}, \quad n \to \infty, \tag{3.7}$$

with a symmetric function $r_0^{(n)}$ and $n \in \mathbb{N}_0$. Evidently, the latter fact implies that the initial condition $k_{0,\varepsilon}^{(n)}$ is singular with respect to $\varepsilon > 0$, which can be interpreted as a growth of the initial densities in $\varepsilon > 0$. The functions $r_0^{(n)}$ are a subject of choice for a particular model. The important case is

$$r_0^{(n)}(x_1, \dots, x_n) = \rho_0(x_1) \cdots \rho_0(x_n),$$

which represents the Poisson statistics. Denote by $L_\varepsilon^{\triangle}$ the operator corresponding to L_ε and by $k_{t,\varepsilon}$ the collection of correlation functions, defined as the solutions of the equation

$$\frac{\partial}{\partial t} k_{t,\varepsilon}(\eta) = L_\varepsilon^{\triangle} k_{t,\varepsilon}(\eta), \quad k_{t,\varepsilon}|_{t=0} = k_{0,\varepsilon}. \tag{3.8}$$

In the final step, we will seek for a limiting correlation function, describing the scaled system, i.e., we want

$$\varepsilon^n k_{t,\varepsilon}^{(n)} \to r_t^{(n)}, \quad n \to \infty, \tag{3.9}$$

for each $n \in \mathbb{N}_0$. Again the collection $r_t = (r_t^{(n)})_{n\in\mathbb{N}_0}$ will satisfy some system of equations similar to (3.5),

$$\frac{\partial}{\partial t} r_t(\eta) = V r_t(\eta), \quad r_t|_{t=0} = r_0. \tag{3.10}$$

This limiting description is known as the Vlasov hierarchy containing less information as the original model, but it is simpler to analyze. It is easy to see that informally, operator V can be obtained as the limit of the "renormalized" operator,

$$L_{\varepsilon,\mathrm{ren}}^{\triangle} := R_\varepsilon L_\varepsilon^{\triangle} R_{\varepsilon^{-1}}, \tag{3.11}$$

where $(R_\varepsilon r)(\eta) := \varepsilon^n r(\eta)$.

Starting from the initial function r_0 as the correlation function of the Poisson measure π_{ρ_0}, one gets that the solution r_t will be of the form

$$r_t^{(n)}(x_1, \dots, x_n) = \rho_t(x_1) \cdots \rho_t(x_n),$$

and so r_t is again the correlation function of a Poisson measure with the new density ρ_t. This property is known as the chaos preservation principle and the density ρ_t is determined by the kinetic (mesoscopic) equation

$$\frac{\partial}{\partial t} \rho_t(x) = v(\rho_t)(x), \quad \rho_t|_{t=0} = \rho_0. \tag{3.12}$$

Now, given a microscopic model through its (formal) Markov generator L, we will say that (3.12) is the kinetic description of the microscopic model. This description is produced by taking the formal limit within (3.9). Without further investigation, it is not known whether also the corresponding solutions converge, i.e., (3.9) holds. We will say that the kinetic description (3.12) corresponds to the microscopic model if (3.8) holds. In particular, this implies that for given initial condition ρ_0 and ρ_t, we have

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$$\varepsilon^n k^{(1)}_{t,\varepsilon} \to \rho_t.$$

The precise notion of the convergence might depend on the particular model and shall be checked for each model separately. In many cases, one knows that ρ_t will be bounded and hence the limit is uniform in all spatial variables. All steps described in this section will be realized now for the analogs of the Kuramoto and Cucker-Smale models for infinite systems of interacting particles postponing the problem of convergence to our future works.

IV. KURAMOTO DYNAMICS FOR INFINITE INTERACTING PARTICLE SYSTEMS

In this section, we present the statistical dynamics of the infinite particle counterpart for the Kuramoto model on configuration space $\Gamma = \Gamma(\mathbb{R} \times \mathbb{R})$ and derive the corresponding kinetic equation through appropriate scaling and chaos preservation assumption. Recall that the classical Kuramoto model is described by the dynamics of phases of $N \in \mathbb{N}$ oscillators, governed by the following ODE system:

$$\frac{d\theta_i}{dt} = \nu_i + \frac{\kappa}{N}\sum_{j\neq i} \sin(\theta_j - \theta_i), \quad 1 \le i \le N, \tag{4.1}$$

where $\theta_i, \nu_i \in \mathbb{R}$ are the phase and natural frequency of the ith oscillator and $\kappa > 0$. Note that contrary to (1.1), the above system of equation does not contain the term with index $j = i$ in the sum at the right-hand side of (4.1). It is essentially the same because the term with index i is nothing but 0.

Let $X^\xi_t = \{z_1(t), \dots, z_N(t)\} \in \Gamma^{(N)}_{0,\mathbb{R}\times\mathbb{R}}$, $t \in \mathbb{R}_+$, be the process with initial random configuration $\xi : \Omega \to \Gamma^{(N)}_{0,\mathbb{R}\times\mathbb{R}}$ distributed by $\mu_0 \in \mathbb{M}^1(\Gamma)$ such that each ith particle $z_i(t) = (\nu_i, \theta_i(t)) \in \mathbb{R} \times \mathbb{R}$, $1 \le i \le N$, evolves according to (4.1). Then, we formally have [cf. with (3.2)]

$$\left(\frac{\partial}{\partial t}\int_\Omega F(X^\xi_t)d\mathbb{P}\right) = \left(\int_\Omega LF(X^\xi_t)d\mathbb{P}\right), \quad F : \Gamma^{(N)}_{0,\mathbb{R}\times\mathbb{R}} \to \mathbb{R}, \tag{4.2}$$

or, equivalently,

$$\frac{\partial}{\partial t}\langle F, \mu_t\rangle = \langle LF, \mu_t\rangle, \quad \mu_t|_{t=0} = \mu_0, \tag{4.3}$$

where μ_t is the distribution of X^ξ_t. It is easy to see from Liouville's theorem that the form (action) of the corresponding evolution operator L (informally dual to the Liouville operator) is given by

$$(LF)(\eta) = \sum_{z\in\eta}\left(\nu_z + \frac{\kappa}{|\eta|}\sum_{y\in\eta\setminus z}\sin(\theta_y - \theta_z)\right)\frac{\partial}{\partial\theta_z}F(\eta), \quad \eta = \{(\nu_\cdot, \theta_\cdot)\} \in \Gamma^{(N)}_{0,\mathbb{R}\times\mathbb{R}}. \tag{4.4}$$

The notation $\eta = \{(\nu_\cdot, \theta_\cdot)\}$ means that any $x \in \eta$ has the form $x = (\nu_x, \theta_x)$, where ν_x stands for the natural frequency of the oscillator x and θ_x for its phase. It is very important to emphasize that we do not specify precisely the domain of this operator at the moment. Of course, in order to give at least pointwise meaning to (4.4), we consider only those F for which the right hand side of (4.4) is finite. Such classes of functions will be discussed later.

Using the abovementioned observation about the form of L, identity (4.3), and general approach presented in Sec. III, we define the statistical analog of the Kuramoto model for infinite particle systems as the Markov process on $\Gamma = \Gamma(\mathbb{R} \times \mathbb{R})$ whose mechanism of evolution is given by the heuristic generator

$$(LF)(\gamma) = \sum_{z\in\gamma}\left(\nu_z + \kappa\sum_{y\in\gamma\setminus z}\sin(\theta_y - \theta_z)\right)\frac{\partial}{\partial\theta_z}F(\gamma), \quad \gamma = \{(\nu_\cdot, \theta_\cdot)\} \in \Gamma. \tag{4.5}$$

Heuristically, for a small time interval Δt, the differential of the observable for the statistical counterpart of the Kuramoto model can be formally calculated as

$$\begin{aligned}\Delta F = F(\gamma + \Delta\gamma) - F(\gamma) &\approx \sum_{z\in\gamma}\left(\frac{\partial}{\partial\theta_z}F(\gamma)(\Delta\theta_z)\right)\\ &= \left[\sum_{z\in\gamma}\left(\nu_z + \kappa\sum_{y\in\gamma\setminus z}\sin(\theta_y - \theta_z)\right)\frac{\partial}{\partial\theta_z}F(\gamma)\right]\Delta t,\end{aligned} \tag{4.6}$$

where we used dynamics (4.1) without the normalizing factor $\frac{1}{N}$. Note that (4.5) does not contain the normalization factor $\frac{1}{|\eta|}$ which is present in (4.4) for a system with finite particles. Nevertheless, the similarity between the classical Kuramoto model for finite systems

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and the statistical model for infinite particle systems given by (4.5) can be seen on the level of evolution operators if we consider spatial mean-field rescaling, which corresponds to the Vlasov (mesoscopic) description for the model given by (4.5), restricted to finite particles only. Clearly, definition (4.5) is technically related to the informal substitution of the finite configuration $\eta \in \Gamma_{0,\mathbb{R}\times\mathbb{R}}$ in (4.4) by the infinite configuration $y \in \Gamma$ and vanishing of the normalization factor $\frac{1}{|\eta|}$ for the part with interaction in (4.5). It is worth pointing out that (4.5) is formally a Markov pregenerator since it shifts constant observables into zero and satisfies the maximum principle for Markov generators.

All technical steps of Sec. III required to get the corresponding kinetic equation will be now realized for the system described by (3.1) with L given by (4.5). Throughout this section, Γ_0 stands always for $\Gamma_{0,\mathbb{R}\times\mathbb{R}}$. In the sequel, the appearance of symbols y, v_y, θ_y in the same formula means that $y = \{(v_y, \theta_y)\}$.

A. Evolution operator for quasiobservables

In this subsection, we obtain the precise form of the evolution operator $\hat{L}$ in Eq. (3.6).

Lemma 4.1. The operator $\hat{L}$ corresponding to L is given by

$$\begin{aligned}(\hat{L}G)(\eta) = (K^{-1}LKG)(\eta) &= \sum_{z\in\eta} v_z \frac{\partial}{\partial\theta_z} G(\eta) + \kappa \sum_{z\in\eta}\sum_{y\in\eta\setminus z} \sin(\theta_y - \theta_z)\frac{\partial}{\partial\theta_z} G(\eta) \\ &+ \kappa \sum_{z\in\eta}\sum_{y\in\eta\setminus z} \sin(\theta_y - \theta_z)\frac{\partial}{\partial\theta_z} G(\eta\setminus y), \quad \eta = \{(v_\cdot, \theta_\cdot)\} \in \Gamma_0\end{aligned} \tag{4.7}$$

for all functions $G \in B_{bs}(\Gamma_0)$ which are differentiable with respect to the variable θ.

Proof. It is clear that $(LKG)(\eta)$ is well defined at least for all $\eta \in \Gamma_0$. Therefore, the expression $K^{-1}LKG$ may be defined via (2.5) pointwisely. It is a simple matter to check that the first summand on the right-hand side of (4.7) is the K-image of the first summand (with v_z) of L given by (4.5). Hence, we immediately proceed to the remaining term of L in (4.5). Note that $\frac{\partial}{\partial\theta_z} G(\xi) = 0$ if $z \notin \xi$. By the definition of the K-transform for $\eta \in \Gamma_0$, we have

$$\begin{aligned}&\kappa \sum_{z\in\eta}\sum_{y\in\eta\setminus z} \sin(\theta_y - \theta_z)\frac{\partial}{\partial\theta_z} KG(\eta) \\ &= \kappa \sum_{z\in\eta}\sum_{y\in\eta\setminus z} \sin(\theta_y - \theta_z)\frac{\partial}{\partial\theta_z} \sum_{\xi\in\eta} G(\xi) \\ &= \kappa \sum_{z\in\eta}\sum_{y\in\eta\setminus z} \sin(\theta_y - \theta_z)\frac{\partial}{\partial\theta_z} \sum_{\substack{\xi\in\eta \\ y,z\in\xi}} G(\xi) + \kappa \sum_{z\in\eta}\sum_{y\in\eta\setminus z} \sin(\theta_y - \theta_z)\frac{\partial}{\partial\theta_z} \sum_{\substack{\xi\in\eta\setminus y \\ z\in\xi}} G(\xi) \\ &= \kappa \sum_{z\in\eta}\sum_{y\in\eta\setminus z} \sin(\theta_y - \theta_z)\frac{\partial}{\partial\theta_z} \sum_{\xi\in(\eta\setminus z)\setminus y} G(\xi\cup z\cup y) \\ &\quad + \kappa \sum_{z\in\eta}\sum_{y\in\eta\setminus z} \sin(\theta_y - \theta_z)\frac{\partial}{\partial\theta_z} \sum_{\xi\in(\eta\setminus z)\setminus y} G(\xi\cup z) \\ &= \kappa \sum_{z\in\eta}\sum_{\xi\in\eta\setminus z}\sum_{y\in\xi} \sin(\theta_y - \theta_z)\frac{\partial}{\partial\theta_z} G(\xi\cup z) \\ &\quad + \kappa \sum_{z\in\eta}\sum_{\xi\in\eta\setminus z}\sum_{y\in\xi} \sin(\theta_y - \theta_z)\frac{\partial}{\partial\theta_z} G((\xi\setminus y)\cup z) \\ &= \kappa \sum_{\xi\in\eta}\sum_{z\in\xi}\sum_{y\in\xi\setminus z} \sin(\theta_y - \theta_z)\frac{\partial}{\partial\theta_z} G(\xi) + \kappa \sum_{\xi\in\eta}\sum_{z\in\xi}\sum_{y\in\xi\setminus z} \sin(\theta_y - \theta_z)\frac{\partial}{\partial\theta_z} G(\xi\setminus y) \\ &= K\left[\kappa \sum_{z\in\cdot}\sum_{y\in\cdot\setminus z} \sin(\theta_y - \theta_z)\frac{\partial}{\partial\theta_z} G(\cdot) + \kappa \sum_{z\in\cdot}\sum_{y\in\cdot\setminus z} \sin(\theta_y - \theta_z)\frac{\partial}{\partial\theta_z} G(\cdot\setminus y)\right](\eta).\end{aligned}$$

It is worth pointing out that all sums in the abovementioned calculation are finite, and hence, they are well-defined. Therefore, after applying the inverse to K-transform, we obtain the desired result. □

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B. Evolution operator for statistical dynamics

Next, we calculate the operator for the dynamics of correlation functions, i.e., the operator $L^{\triangle}$ introduced in (3.5).

Lemma 4.2. The operator $L^{\triangle}$ is given by

$$(L^{\triangle}k)(\eta) = -\sum_{z\in\eta} v_z \frac{\partial}{\partial\theta_z}k(\eta) + \kappa\sum_{z\in\eta}\sum_{y\in\eta\setminus z}\left[\cos(\theta_y-\theta_z)k(\eta) - \sin(\theta_y-\theta_z)\frac{\partial}{\partial\theta_z}k(\eta)\right]$$

$$+\kappa\int_{\mathbb{R}\times\mathbb{R}}\sum_{z\in\eta}\left[\cos(\theta_y-\theta_z)k(\eta\cup y) - \sin(\theta_y-\theta_z)\frac{\partial}{\partial\theta_z}k(\eta\cup y)\right]dv_y d\theta_y,\ \eta = \{(v_\cdot,\theta_\cdot)\}\in\Gamma_0$$

for any function k on Γ_0 which is differentiable with respect to θ and for which the right-hand side of the above formula makes sense.

Remark 4.1. Lemma 4.2 gives only the formal calculation of the action of $L^{\triangle}$. Therefore, for its proof, we may consider only those $k:\Gamma_0\to\mathbb{R}$ for which $\langle\langle \hat{L}G, k\rangle\rangle$ and $\langle\langle G, L^{\triangle}k\rangle\rangle$ have sense for all $G\in B_{bs}(\Gamma_0)$.

Proof. According to Lemma 4.1 for all $G\in B_{bs}(\Gamma_0)$, we have

$$\begin{aligned}\int_{\Gamma_0}\hat{L}G(\eta)k(\eta)\lambda(d\eta) &= \int_{\Gamma_0}\Bigg[\sum_{z\in\eta}v_z\frac{\partial}{\partial\theta_z}G(\eta) + \sum_{z\in\eta}\sum_{y\in\eta\setminus z}\sin(\theta_y-\theta_z)\kappa\frac{\partial}{\partial\theta_z}G(\eta)\\ &\quad+\sum_{z\in\eta}\sum_{y\in\eta\setminus z}\sin(\theta_y-\theta_z)\kappa\frac{\partial}{\partial\theta_z}G(\eta\setminus y)\Bigg]k(\eta)\lambda(d\eta)\\ &=: \mathcal{I}_{11}+\mathcal{I}_{12}+\mathcal{I}_{13}.\end{aligned}$$

Below, we calculate terms $\mathcal{I}_{1i}$, $i=1,2,3$, separately. To this end, it is very important to emphasize that $(\hat{L}G)(\emptyset)=0$. For abbreviation, we set $\bar{z}^n=(z_1,\dots,z_n)$ and $d\bar{z}^n = dv_1d\theta_1\dots dv_nd\theta_n$ for all $n\geq 1$.

- Calculation of $\mathcal{I}_{11}$: Using integration by parts, the definition of the Lebesgue-Poisson measure, and the fact that G has bounded support, we can assert that

$$\begin{aligned}\mathcal{I}_{11} &= \int_{\Gamma_0}\sum_{z\in\eta}v_z\left[\frac{\partial}{\partial\theta_z}G(\eta)\right]k(\eta)\lambda(d\eta) = \sum_{n=1}^{\infty}\frac{1}{n!}\int_{(\mathbb{R}\times\mathbb{R})^n}\sum_{i=1}^{n}v_{z_i}\frac{\partial G^{(n)}}{\partial\theta_{z_i}}(\bar{z}^n)k^{(n)}(\bar{z}^n)d\bar{z}^n\\ &= -\sum_{n=1}^{\infty}\frac{1}{n!}\int_{(\mathbb{R}\times\mathbb{R})^n}\sum_{i=1}^{n}v_{z_i}G^{(n)}(\bar{z}^n)\frac{\partial k^{(n)}}{\partial\theta_{z_i}}(\bar{z}^n)d\bar{z}^n = -\int_{\Gamma_0}\sum_{z\in\eta}v_zG(\eta)\frac{\partial}{\partial\theta_z}k(\eta)\lambda(d\eta).\end{aligned}$$

- Calculation of $\mathcal{I}_{12}$: Similar to the case of $\mathcal{I}_{11}$, we get

$$\begin{aligned}\mathcal{I}_{12} &= \int_{\Gamma_0}\sum_{z\in\eta}\sum_{y\in\eta\setminus z}\sin(\theta_y-\theta_z)\kappa\left[\frac{\partial}{\partial\theta_z}G(\eta)\right]k(\eta)\lambda(d\eta)\\ &= \sum_{n=1}^{\infty}\sum_{i=1}^{n}\sum_{j\neq i}\frac{1}{n!}\kappa\int_{(\mathbb{R}\times\mathbb{R})^n}\left[\frac{\partial G^{(n)}}{\partial\theta_{z_i}}(\bar{z}^n)\right]\sin(\theta_{z_j}-\theta_{z_i})k^{(n)}(\bar{z}^n)d\bar{z}^n.\end{aligned}$$

For the function $g(\bar{z}^n)=\sin(\theta_{z_j}-\theta_{z_i})k^{(n)}(\bar{z}^n)$, we have

$$\frac{\partial g}{\partial\theta_{z_i}}(\bar{z}^n) = -\cos(\theta_{z_j}-\theta_{z_i})k^{(n)}(\bar{z}^n) + \sin(\theta_{z_j}-\theta_{z_i})\frac{\partial k^{(n)}}{\partial\theta_{z_i}}(\bar{z}^n).$$

Consequently,

$$\mathcal{I}_{12} = \int_{\Gamma_0}G(\eta)\sum_{z\in\eta}\sum_{y\in\eta\setminus z}\kappa\left[\cos(\theta_y-\theta_z)k(\eta) - \sin(\theta_y-\theta_z)\frac{\partial}{\partial\theta_z}k(\eta)\right]\lambda(d\eta).$$

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- Calculation of $\mathcal{I}_{13}$: Since we only consider k that the second and third terms are well-defined, we may use Lemma 2.1. Thus, we again use integration by parts and Lemma 2.1 to obtain

$$\begin{aligned}\mathcal{I}_{13} &= \int_{\Gamma_0}\sum_{z\in\eta}\sum_{y\in\eta\setminus z}\sin(\theta_y-\theta_z)\kappa\Big[\frac{\partial}{\partial\theta_z}G(\eta\setminus y)\Big]k(\eta)\lambda(d\eta)\\ &= -\int_{\Gamma_0}\sum_{z\in\eta}\sum_{y\in\eta\setminus z}G(\eta\setminus y)\kappa\Big[-\cos(\theta_y-\theta_z)k(\eta)+\sin(\theta_y-\theta_z)\frac{\partial}{\partial\theta_z}k(\eta)\Big]\lambda(d\eta)\\ &= \int_{\Gamma_0}\int_{\mathbb{R}\times\mathbb{R}}\sum_{z\in\eta}G(\eta)\kappa\Big[\cos(\theta_y-\theta_z)k(\eta\cup y)-\sin(\theta_y-\theta_z)\frac{\partial}{\partial\theta_z}k(\eta\cup y)\Big]dv_y\,d\theta_y\lambda(d\eta),\end{aligned}$$

  where $y=\{(v_y,\theta_y)\}$.

Finally, we combine all calculations of $\mathcal{I}_{1i}$, $i=1,2,3$, to obtain the desired statement. □

Remark 4.2. It is easily seen that $L^{\triangle}$ is not well-defined for $k_C(\eta)=C^{|\eta|}$ (the correlation function of the Poisson measure with intensity C). This is, of course, related to the fact that $\cos(\theta_i-\theta_j)$ is not integrable on $\mathbb{R}\times\mathbb{R}$. Hence, the Vlasov scaling scheme proposed in Sec. III seems to be realized only for chaotic Poisson states with density being integrable functions. This technical problem appears for the statistical analog of the Kuramoto model for infinite interacting particle systems since the framework of the Vlasov scaling scheme proposed in Sec. III is related to the strong forms of Eqs. (3.5), (3.8), and (3.10). However, this problem can be avoided if k_C is regarded as distribution over $B_{bs}(\Gamma_0)$. The latter situation requires a separate justification of the scheme proposed in Sec. III. On the other hand, the case of integrable correlation functions yields that effectively our stochastic dynamics evolves through finite configurations only. Note that the case of an integrable first order correlation function is referred to the zero density case in statistical physics. The Vlasov equation for integrable functions was investigated, e.g., in detail by Kozlov.[22]

C. Vlasov scaling for Kuramoto

In this subsection, we follow the scheme proposed in Sec. III (cf. Ref. 10) to obtain the Vlasov equation for the Kuramoto model. Below, we scale the operator $L^{\triangle}$ and derive the renormalized version of it. To do this, we scale the interaction potential $\sin(\theta_y-\theta_z)$ in the representation (4.5) of the operator L multiplying it with $\varepsilon>0$. Let L_ε be the corresponding operator after such a scaling, and let $L^{\triangle}_\varepsilon$ be the scaling of $L^{\triangle}$ corresponding to L_ε. Effectively it means for $L^{\triangle}$ that we scale communications $\sin\mapsto\varepsilon\sin$ and $\cos\mapsto\varepsilon\cos$, respectively. Namely,

$$\begin{aligned}(L^{\triangle}_\varepsilon k)(\eta) &= -\sum_{z\in\eta}v_z\frac{\partial}{\partial\theta_z}k(\eta)+\varepsilon\kappa\sum_{z\in\eta}\sum_{y\in\eta\setminus z}\Big[\cos(\theta_y-\theta_z)k(\eta)-\sin(\theta_y-\theta_z)\frac{\partial}{\partial\theta_z}k(\eta)\Big]\\ &\quad+\varepsilon\kappa\int_{\mathbb{R}\times\mathbb{R}}\sum_{z\in\eta}\Big[\cos(\theta_y-\theta_z)k(\eta\cup y)-\sin(\theta_y-\theta_z)\frac{\partial}{\partial\theta_z}k(\eta\cup y)\Big]dy.\end{aligned}$$

For the renormalized operator $L^{\triangle}_{\varepsilon,\mathrm{ren}}=R_\varepsilon L^{\triangle}_\varepsilon R_{\varepsilon^{-1}}$, we get the following formula:

$$\begin{aligned}(L^{\triangle}_{\varepsilon,\mathrm{ren}}k)(\eta) &= -\sum_{z\in\eta}v_z\frac{\partial}{\partial\theta_z}k(\eta)+\kappa\sum_{z\in\eta}\sum_{y\in\eta\setminus z}\varepsilon\Big[\cos(\theta_y-\theta_z)k(\eta)-\sin(\theta_y-\theta_z)\frac{\partial}{\partial\theta_z}k(\eta)\Big]\\ &\quad+\kappa\int_{\mathbb{R}\times\mathbb{R}}\sum_{z\in\eta}\Big[\cos(\theta_y-\theta_z)k(\eta\cup y)-\sin(\theta_y-\theta_z)\frac{\partial}{\partial\theta_z}k(\eta\cup y)\Big]dy,\end{aligned}$$

which converges pointwisely as $\varepsilon\to0$ to

$$\begin{aligned}(Vk)(\eta) &= -\sum_{z\in\eta}v_z\frac{\partial}{\partial\theta_z}k(\eta)\\ &\quad+\kappa\int_{\mathbb{R}\times\mathbb{R}}\sum_{z\in\eta}\Big[\cos(\theta_y-\theta_z)k(\eta\cup y)-\sin(\theta_y-\theta_z)\frac{\partial}{\partial\theta_z}k(\eta\cup y)\Big]dy.\end{aligned}$$

Remark 4.3. It is easily seen, analogously as for the operator $L^{\triangle}$, that the operator V is not well-defined for $k_C(\eta)=C^{|\eta|}$ (cf. Remark 4.2).

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Now, to obtain the mesoscopic equation, we assume that there exists a solution to the following Vlasov hierarchy:

$$\frac{\partial}{\partial t}k_t(\eta) = Vk_t(\eta), \quad k_t|_{t=0} = k_0, \tag{4.8}$$

which obeys the chaos preservation property, i.e., there exists a function $f : \mathbb{R} \times \mathbb{R} \times [0, \infty) \to \mathbb{R}$ such that

$$k_0(\eta) = \prod_{x\in\eta} f_0(x) \Rightarrow k_t(\eta) = \prod_{x\in\eta} f_t(x). \tag{4.9}$$

Here, we also informally assume that the product-function $\prod_{x\in\eta} f_t(x)$ is included in the domain of the operator V. Under this setting, our goal is to use (4.8) and (4.9) to find the operator v such that

$$\frac{\partial}{\partial t}f_t(x) = v(f_t)(x), \quad x \in \mathbb{R} \times \mathbb{R}. \tag{4.10}$$

Theorem 4.1. *Suppose the Vlasov hierarchy equation (4.8) has a solution satisfying the chaos preservation property (4.9) for some function f on $\mathbb{R} \times \mathbb{R} \times [0, \infty)$ which is differentiable with respect to θ and for which the right-hand side of (4.11) makes sense (cf. Remarks 4.2 and 4.3). Then, the evolution operator in (4.10) is given by*

$$v(f)(x) = -v_x \frac{\partial}{\partial\theta_x} f(x) - \kappa \frac{\partial}{\partial\theta_x} \int_{\mathbb{R}\times\mathbb{R}} \sin(\theta_y - \theta_x) f(x) f(y) dy. \tag{4.11}$$

Proof. We use the chaos preservation property to rewrite the left-hand side of (4.8) as

$$\frac{\partial}{\partial t}k_t(\eta) = \frac{\partial}{\partial t}\left(\prod_{x\in\eta} f_t(x)\right) = \sum_{x\in\gamma}\left(\prod_{y\in\gamma\setminus x} f_t(y)\right)\frac{\partial}{\partial t}f_t(x).$$

Hence, we would like to isolate the factor

$$\sum_{x\in\gamma}\left(\prod_{y\in\gamma\setminus x} f_t(y)\right)$$

from the right-hand side of Eq. (4.8). The remainder will be the operator v,

$$\begin{aligned}
(Vk_t)(\eta) &= -\sum_{x\in\eta} v_x \frac{\partial}{\partial\theta_x}\left(\prod_{z\in\eta} f_t(z)\right) \\
&\quad + \kappa \int_{\mathbb{R}\times\mathbb{R}} \sum_{x\in\eta}\left[\cos(\theta_y - \theta_x)\prod_{z\in\eta} f_t(z) f_t(y) - \sin(\theta_y - \theta_x)\frac{\partial}{\partial\theta_x}\prod_{z\in\eta} f_t(z) f_t(y)\right] dy \\
&= -\sum_{x\in\eta}\prod_{z\in\eta\setminus x} f_t(z) v_x \frac{\partial}{\partial\theta_x} f_t(x) \\
&\quad + \sum_{x\in\eta}\prod_{z\in\eta\setminus x} f_t(z)\kappa \int_{\mathbb{R}\times\mathbb{R}} \cos(\theta_y - \theta_x) f_t(x) f_t(y) - \sin(\theta_y - \theta_x)\frac{\partial}{\partial\theta_x} f_t(x) f_t(y) dy.
\end{aligned}$$

Hence, for fixed x, we obtain

$$\begin{aligned}
v(f_t)(x) &= -v_x \frac{\partial}{\partial\theta_x} f_t(x) + \kappa \int_{\mathbb{R}\times\mathbb{R}} \cos(\theta_y - \theta_x) f_t(x) f_t(y) - \sin(\theta_y - \theta_x)\frac{\partial}{\partial\theta_x} f_t(x) f_t(y) dy \\
&= -v_x \frac{\partial}{\partial\theta_x} f_t(x) - \kappa \frac{\partial}{\partial\theta_x} \int_{\mathbb{R}\times\mathbb{R}} \sin(\theta_y - \theta_x) f_t(x) f_t(y) dy,
\end{aligned}$$

which is the desired conclusion. □

Remark 4.4. Let f_t be a solution to (4.10) with v given in Theorem 4.1, and assume that $k_t(\eta) = \prod_{x\in\eta} f_t(x)$ belongs to the domain of the operator V. Then, k_t is a solution to the Vlasov hierarchy equation (4.8).

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As we can see, the mesoscopic equation (4.10) with the operator v calculated in Theorem 4.1 coincides with the Kuramoto-Sakaguchi equation (1.3).

V. C-S DYNAMICS FOR INFINITE INTERACTING PARTICLE SYSTEMS

In this section, we write Γ and Γ_0 for $\Gamma(\mathbb{R}^d \times \mathbb{R}^d)$ and $\Gamma_{0,\mathbb{R}^d\times\mathbb{R}^d}$, respectively. The statistical analog of the C-S model for infinite interacting particle systems can be introduced as follows. We recall the heuristic derivation of the evolution operator for the Kuramoto model in (4.4) and use the C-S model (1.2) for finite particle systems to define the statistical counterpart of the C-S model for infinite interacting particle systems. Namely, acting in the same fashion as for the Kuramoto model, we define the corresponding mechanism of the evolution for observables for the statistical counterpart of C-S by

$$LF(\gamma) := \sum_{x\in\gamma} v_x \cdot \nabla_{r_x} F(\gamma) + \sum_{y\in\gamma\setminus x} \phi(r_x - r_y)(v_y - v_x)\cdot\nabla_{v_x} F(\gamma), \quad \gamma = \{(r_\cdot, v_\cdot)\} \in \Gamma. \tag{5.1}$$

As in Sec. IV, the notation $\gamma = \{(r_\cdot, v_\cdot)\}$ means that any $x \in \gamma$ is actually a pair $x = (r_x, v_x)$, which is composed of the position and velocity vectors. From now on, the appearance of symbols y, r_y, v_y in the same formula indicates that $y = \{(r_y, v_y)\}$.

We derive the kinetic equation for the C-S model using the abstract scheme described in Sec. III. According to that, the first step is to obtain the form (action) of the operator $\hat{L}$.

Lemma 5.1. The operator $\hat{L}$ corresponding to L is given by

$$\begin{aligned}(\hat{L}G)(\eta) = &\sum_{z\in\eta} v_z \cdot \nabla_{r_z} G(\eta) + \sum_{z\in\eta}\sum_{y\in\eta\setminus z} \phi(x_z - x_y)(v_y - v_z)\cdot\nabla_{v_z} G(\eta)\\ &+ \sum_{z\in\eta}\sum_{y\in\eta\setminus z} \phi(x_z - x_y)(v_y - v_z)\cdot\nabla_{v_z} G(\eta\setminus y), \quad \eta = \{(r_\cdot, v_\cdot)\} \in \Gamma_0\end{aligned} \tag{5.2}$$

for any $G \in B_{bs}(\Gamma_0)$ which is differentiable with respect to r and v and for which the right-hand side of the above formula makes sense.

Proof. For any $G \in B_{bs}(\Gamma_0)$ and $\eta \in \Gamma_0$, we use the definition of the K-transform to have

$$\begin{aligned}(LKG)(\eta) &= \sum_{x\in\eta} v_x \cdot \nabla_{r_x} KG(\eta) + \sum_{x\in\eta}\sum_{y\in\eta\setminus x} \phi(r_x - r_y)(v_y - v_x)\cdot\nabla_{v_x} KG(\eta)\\ &= \sum_{x\in\eta}\sum_{\zeta\Subset\eta, x\in\zeta} v_x \cdot \nabla_{r_x} G(\zeta) + \sum_{x\in\eta}\sum_{y\in\eta\setminus x}\sum_{\zeta\Subset\eta, x\in\zeta} \phi(r_x - r_y)(v_y - v_x)\cdot\nabla_{v_x} G(\zeta)\\ &= \sum_{x\in\eta}\sum_{\zeta\Subset\eta\setminus x} v_x \cdot \nabla_{r_x} G(\zeta\cup x) + \sum_{x\in\eta}\sum_{y\in\eta\setminus x}\sum_{\zeta\Subset\eta\setminus x} \phi(r_x - r_y)(v_y - v_x)\cdot\nabla_{v_x} G(\zeta\cup x)\\ &=: \sum_{\zeta\Subset\eta}\sum_{x\in\zeta} v_x \cdot \nabla_{r_x} G(\zeta) + \mathcal{I} = K\sum_{x\in\cdot} (v_x \cdot \nabla_{r_x} G(\cdot))(\eta) + \mathcal{I},\end{aligned}$$

where

$$\begin{aligned}\mathcal{I} &= \sum_{x\in\eta}\sum_{y\in\eta\setminus x}\sum_{\zeta\Subset\eta\setminus x} \phi(r_x - r_y)(v_y - v_x)\cdot\nabla_{v_x} G(\zeta\cup x)\\ &= \sum_{x\in\eta}\sum_{y\in\eta\setminus x}\sum_{\zeta\Subset\eta\setminus\{x,y\}} \phi(r_x - r_y)(v_y - v_x)\cdot\nabla_{v_x} G(\zeta\cup x)\\ &\quad + \sum_{x\in\eta}\sum_{y\in\eta\setminus x}\sum_{\zeta\Subset\eta\setminus x,\, y\in\zeta} \phi(r_x - r_y)(v_y - v_x)\cdot\nabla_{v_x} G(\zeta\cup x)\\ &= \sum_{\zeta\Subset\eta}\sum_{x\in\zeta}\sum_{y\in\zeta\setminus x} \phi(r_x - r_y)(v_y - v_x)\cdot\nabla_{v_x} G(\zeta\setminus y)\\ &\quad + \sum_{x\in\eta}\sum_{y\in\eta\setminus x}\sum_{\zeta\Subset\eta\setminus x,\, y\in\zeta} \phi(r_x - r_y)(v_y - v_x)\cdot\nabla_{v_x} G(\zeta\cup x)\\ &= \sum_{\zeta\Subset\eta}\sum_{x\in\zeta}\sum_{y\in\zeta\setminus x} \phi(r_x - r_y)(v_y - v_x)\cdot[\nabla_{v_x} G(\zeta\setminus y) + \nabla_{v_x} G(\zeta)]\\ &= K\left[\sum_{x\in\cdot}\sum_{y\in\cdot\setminus x} \phi(r_x - r_y)(v_y - v_x)\cdot[\nabla_{v_x} G(\cdot\setminus y) + \nabla_{v_x} G(\cdot)]\right](\eta).\end{aligned}$$

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This gives the exact formula (5.2) for the operator $\hat{L} = K^{-1}LK$. □

Again, we use the natural dual relation between the space of quasiobservables and correlation functions (3.4) to obtain the dual operator $L^{\triangle}$ of $\hat{L}$.

Lemma 5.2. The operator $L^{\triangle}$ is given by

$$(L^{\triangle}k)(\eta) = -\sum_{x\in\eta} v_x \cdot \nabla_{r_x} k(\eta) + \sum_{x\in\eta}\sum_{y\in\eta\setminus x} \phi(r_x - r_y)(dk(\eta) - (v_y - v_x)\cdot\nabla_{v_x}k(\eta))$$

$$+ \int_{\mathbb{R}^d\times\mathbb{R}^d} \sum_{x\in\eta} \phi(r_x - r_y)(dk(\eta\cup y) - (v_y - v_x)\cdot\nabla_{v_x}k(\eta\cup y))dr_y dv_y,\ \eta = \{(r.,\ v.)\} \in \eta_0$$

for any function k on Γ_0 which is differentiable with respect to r and v and for which the right-hand side of the above formula makes sense.

Remark 5.1. As in Remark 4.1, for the Proof of Lemma 5.2, we may consider only those $k : \Gamma_0 \to \mathbb{R}$ for which $\langle\langle \hat{L}G,\ k\rangle\rangle$ and $\langle\langle G,\ L^{\triangle}k\rangle\rangle$ have sense for all $G \in B_{bs}(\Gamma_0)$.

Proof. We use the formula for $\hat{L}$ derived in Lemma 5.1 to obtain

$$\begin{aligned}\int_{\Gamma_0} (\hat{L}G)(\eta)k(\eta)\lambda(d\eta) = \int_{\Gamma_0}\Bigg[&\sum_{x\in\eta} v_x\cdot\nabla_{r_x}G(\eta) + \sum_{x\in\eta}\sum_{y\in\eta\setminus x}\phi(r_x - r_y)(v_y - v_x)\cdot\nabla_{v_x}G(\eta)\\ &+ \sum_{x\in\eta}\sum_{y\in\eta\setminus x}\phi(r_x - r_y)(v_y - v_x)\cdot\nabla_{v_x}G(\eta\setminus y)\Bigg]k(\eta)\lambda(d\eta)\\ =:\ &\mathcal{I}_{21} + \mathcal{I}_{22} + \mathcal{I}_{23}.\end{aligned}$$

To shorten the notation in the next calculations, we set $\bar{r}^n := (r_1, \dots, r_n)$, $\bar{v}^n := (v_1, \dots, v_n)$, $d\bar{r}^n := dr_1\cdots dr_n$, and $d\bar{v}^n := dv_1\cdots dv_n$ for all $n \geq 1$.

- Calculation of $\mathcal{I}_{21}$: According to the definition of the Lebesgue-Poisson measure and the fact that $(\hat{L}G)(\emptyset) = 0$, we have

$$\begin{aligned}\mathcal{I}_{21} &:= \int_{\Gamma_0}\sum_{x\in\eta} v_x\cdot\nabla_{r_x}G(\eta)k(\eta)\lambda(d\eta)\\ &= \sum_{n=1}^{\infty}\frac{1}{n!}\int_{\mathbb{R}^{nd}}\int_{\mathbb{R}^{nd}}\sum_{i=1}^{n} v_i\cdot\nabla_{r_i}G^{(n)}(\bar{r}^n,\ \bar{v}^n)k^{(n)}(\bar{r}^n,\ \bar{v}^n)d\bar{r}^n d\bar{v}^n.\end{aligned}$$

  Let $r_i = (r_i^1, \dots, r_i^d)$ and $v_i = (v_i^1, \dots, v_i^d)$, and then,

$$\mathcal{I}_{21} = \sum_{n=1}^{\infty}\sum_{i=1}^{n}\sum_{j=1}^{d}\frac{1}{n!}\int_{\mathbb{R}^{nd}}\int_{\mathbb{R}^{nd}} v_i^j\frac{\partial G^{(n)}}{\partial r_i^j}(\bar{r}^n,\ \bar{v}^n)k^{(n)}(\bar{r}^n,\ \bar{v}^n)d\bar{r}^n d\bar{v}^n.$$

  Now, the formula for integration by parts and the fact that G has bounded support imply

$$\begin{aligned}\mathcal{I}_{21} &= -\sum_{n=1}^{\infty}\sum_{i=1}^{n}\sum_{j=1}^{d}\frac{1}{n!}\int_{\mathbb{R}^{nd}}\int_{\mathbb{R}^{nd}} G^{(n)}(\bar{r}^n,\ \bar{v}^n)v_i^j\frac{\partial k^{(n)}}{\partial r_i^j}(\bar{r}^n,\ \bar{v}^n)d\bar{r}^n d\bar{v}^n\\ &= -\int_{\Gamma_0}\sum_{x\in\eta} G(\eta)(v_x\cdot\nabla_{r_x}k(\eta))\lambda(d\eta).\end{aligned}$$

- Calculation of $\mathcal{I}_{22}$: The definition of the Lebesgue-Poisson measure implies

$$\begin{aligned}\mathcal{I}_{22} &:= \int_{\Gamma_0}\sum_{x\in\eta}\sum_{y\in\eta\setminus x}\phi(r_x - r_y)(v_y - v_x)\cdot\nabla_{v_x}G(\eta)k(\eta)\lambda(d\eta)\\ &= \sum_{n=1}^{\infty}\sum_{i=1}^{n}\sum_{j\neq i}\sum_{l=1}^{d}\frac{1}{n!}\int_{\mathbb{R}^{nd}}\int_{\mathbb{R}^{nd}}\frac{\partial G^{(n)}}{\partial v_i^l}(\bar{r}^n,\ \bar{v}^n)\phi(r_i - r_j)(v_j^l - v_i^l)k^{(n)}(\bar{r}^n,\ \bar{v}^n)d\bar{r}^n d\bar{v}^n.\end{aligned}$$

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On account of the following formula for the function $g(\overline{r}^n, \overline{v}^n) := (v_j^l - v_i^l)k^{(n)}(\overline{r}^n, \overline{v}^n)$,

$$\frac{\partial g}{\partial v_i^l}(\overline{r}^n, \overline{v}^n) = -k^{(n)}(\overline{r}, \overline{v}) + (v_j^l - v_i^l)\frac{\partial k^{(n)}}{\partial v_i^l}(\overline{r}, \overline{v}),$$

it may be concluded from the formula for integration by parts that

$$\mathcal{I}_{22} = \int_{\Gamma_0} \sum_{x\in\eta} \sum_{y\in\eta\setminus x} G(\eta)\phi(r_x - r_y)(dk(\eta) - (v_y - v_x)\cdot\nabla_{v_x}k(\eta))\lambda(d\eta).$$

- Calculation of $\mathcal{I}_{23}$: By abuse of notation, we use the symbols $\overline{r}_j^n$ and $\overline{v}_j^n$, $1 \le j \le n$, to denote $\overline{r}^n$ and $\overline{v}^n$ with the excluded jth coordinate. We can now proceed analogously to the case of $\mathcal{I}_{22}$, obtaining

$$\begin{aligned}\mathcal{I}_{23} &:= \int_{\Gamma_0} \sum_{x\in\eta} \sum_{y\in\eta\setminus x} \phi(r_x - r_y)(v_y - v_x)\cdot\nabla_{v_x}G(\eta\setminus y)k(\eta)\lambda(d\eta)\\ &= \sum_{n=1}^{\infty}\sum_{i=1}^{n}\sum_{j\neq i}\sum_{l=1}^{d}\frac{1}{n!}\int_{\mathbb{R}^{nd}}\int_{\mathbb{R}^{nd}}\frac{\partial G^{(n-1)}}{\partial v_i^l}(\overline{r}_j^n, \overline{v}_j^n)\phi(r_i - r_j)(v_j^l - v_i^l)k^{(n)}(\overline{r}^n, \overline{v}^n)d\overline{r}^n d\overline{v}^n.\end{aligned}$$

The formula for integration by parts, Lemma 2.1, and observation similar to that in the calculation of $\mathcal{I}_{22}$ show

$$\begin{aligned}\mathcal{I}_{23} &= \int_{\Gamma_0} \sum_{x\in\eta} \sum_{y\in\eta\setminus x} G(\eta\setminus y)\phi(r_x - r_y)(dk(\eta) - (v_y - v_x)\cdot\nabla_{v_x}k(\eta))\lambda(d\eta)\\ &= \int_{\Gamma_0}\int_{\mathbb{R}^d\times\mathbb{R}^d}\sum_{x\in\eta} G(\eta)\phi(r_x - r_y)(dk(\eta\cup y) - (v_y - v_x)\cdot\nabla_{v_x}k(\eta\cup y))dr_y dv_y\lambda(d\eta).\end{aligned}$$

Here, again we only consider k so that we may use Lemma 2.1. Combining all calculations for $\mathcal{I}_{2i}$ for $i = 1, 2, 3$, we obtain

$$\begin{aligned}(L^{\triangle}k)(\eta) = &-\sum_{x\in\eta} v_x\cdot\nabla_{r_x}k(\eta) + \sum_{x\in\eta}\sum_{y\in\eta\setminus x}\phi(r_x - r_y)(dk(\eta) - (v_y - v_x)\cdot\nabla_{v_x}k(\eta))\\ &+ \int_{\mathbb{R}^d\times\mathbb{R}^d}\sum_{x\in\eta}\phi(r_x - r_y)(dk(\eta\cup y) - (v_y - v_x)\cdot\nabla_{v_x}k(\eta\cup y))dr_y dv_y,\end{aligned}$$

which completes the proof. □

We now turn to the next step. Namely, the remainder of this section will be devoted to the Vlasov scaling. To this end, we first derive the renormalized version of the operator $L^{\triangle}$. The scaling of $L^{\triangle}$ is defined by scaling the communication weight, i.e., $\phi \mapsto \varepsilon\phi$,

$$\begin{aligned}(L_\varepsilon^{\triangle}k)(\eta) = &-\sum_{x\in\eta} v_x\cdot\nabla_{r_x}k(\eta) + \sum_{x\in\eta}\sum_{y\in\eta\setminus x}\varepsilon\phi(r_x - r_y)(dk(\eta) - (v_y - v_x)\cdot\nabla_{v_x}k(\eta))\\ &+ \int_{\mathbb{R}^d\times\mathbb{R}^d}\sum_{x\in\eta}\varepsilon\phi(r_x - r_y)(dk(\eta\cup y) - (v_y - v_x)\cdot\nabla_{v_x}k(\eta\cup y))dr_y dv_y.\end{aligned}$$

Then, the renormalized operator $L_{\varepsilon,\mathrm{ren}}^{\triangle}$ is given by

$$\begin{aligned}(L_{\varepsilon,\mathrm{ren}}^{\triangle}k)(\eta) = &-\sum_{x\in\eta} v_x\cdot\nabla_{r_x}k(\eta) + \varepsilon\sum_{x\in\eta}\sum_{y\in\eta\setminus x}\phi(r_x - r_y)(dk(\eta) - (v_y - v_x)\cdot\nabla_{v_x}k(\eta))\\ &+ \int_{\mathbb{R}^d\times\mathbb{R}^d}\sum_{x\in\eta}\phi(r_x - r_y)(dk(\eta\cup y) - (v_y - v_x)\cdot\nabla_{v_x}k(\eta\cup y))dr_y dv_y.\end{aligned}$$

It converges pointwisely as $\varepsilon \to 0$ to

$$(Vk)(\eta) = -\sum_{x\in\eta} v_x\cdot\nabla_{r_x}k(\eta) + \int_{\mathbb{R}^d\times\mathbb{R}^d}\sum_{x\in\eta}\phi(r_x - r_y)(dk(\eta\cup y) - (v_y - v_x)\cdot\nabla_{v_x}k(\eta\cup y))dr_y dv_y.$$

Remark 5.2. It is easily seen that operators $L^{\triangle}$ and V are not well-defined for $k_C(\eta) = C^{|\eta|}$ (cf. Remark 4.2). Hence, the Vlasov scaling scheme proposed in Sec. III seems to be realized only for chaotic Poisson states with density being integrable functions. This technical problem appears

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for the analog of the Cucker-Smale model for infinite interacting particle systems since the framework of the Vlasov scaling scheme proposed in Sec. III is related to the strong forms of Eqs. (3.5), (3.8), and (3.10). However, this problem could be avoided if k_C is regarded as distribution over $B_{bs}(\Gamma_0)$. The latter situation requires a separate justification of the scheme proposed in Sec. III.

We again assume that the following Vlasov hierarchy equation has a solution to

$$\frac{\partial}{\partial t}k_t = Vk_t, \quad k_t|_{t=0} = k_0, \tag{5.3}$$

with the chaos-preservation property (4.9). Then, we find the operator v such that the density f_t satisfies the following evolution equation:

$$\frac{df_t}{dt} = v(f_t). \tag{5.4}$$

Theorem 5.1. *Suppose the Vlasov hierarchy equation (5.3) has a solution satisfying the chaos preservation property (4.9) for some function f which is differentiable with respect to r and v and for which the right-hand side of (5.5) makes sense (cf. Remark 5.2). Then, the operator v is given by*

$$v(f)(x) = -v_x \cdot \nabla_{r_x} f(x) - \nabla_{v_x} \cdot \left[\int_{\mathbb{R}^d \times \mathbb{R}^d} \phi(r_x - r_y)(v_y - v_x) f(x) f(y) dr_y dv_y \right]. \tag{5.5}$$

Proof. Similar to the Kuramoto dynamics case, the evolution equation $\frac{dk_t}{dt} = V(k_t)$ can be written as

$$\begin{aligned}\sum_{x\in\eta}\left[\left(\prod_{z\in\eta\setminus x} f_t(z)\right) v(f_t(x))\right] &= -\sum_{x\in\eta}\left(v_x \cdot \nabla_{r_x} f_t(x) \prod_{z\in\eta\setminus x} f_t(z)\right) \\ &- \left(\prod_{z\in\eta\setminus x} f_t(z)\right) \int_{\mathbb{R}^d\times\mathbb{R}^d} \sum_{x\in\eta} \phi(r_x - r_y)[-df_t(x) + (v_y - v_x)\cdot\nabla_{v_x} f_t(x)] f_t(y) dv_y dr_y.\end{aligned}$$

Now, for fixed $x \in \eta$, we compare the coefficients of $\prod_{z\in\eta\setminus x} f_t(z)$ to obtain

$$\begin{aligned} v(f_t)(x) &= -v_x \cdot \nabla_{r_x} f_t(x) + df_t(x) \int_{\mathbb{R}^d\times\mathbb{R}^d} \phi(r_x - r_y) f_t(y) dr_y dv_y \\ &- \nabla_{v_x} f_t(x) \cdot \left(\int_{\mathbb{R}^d\times\mathbb{R}^d} \phi(r_x - r_y)(v_y - v_x) f_t(y) dv_y dr_y \right) \\ &= -v_x \cdot \nabla_{r_x} f_t(x) - \nabla_{v_x} \cdot \left[\int_{\mathbb{R}^d\times\mathbb{R}^d} \phi(r_x - r_y)(v_y - v_x) f_t(x) f_t(y) dr_y dv_y \right], \end{aligned}$$

which establishes the desired formula. □

Remark 5.3. Let f_t be a solution to (5.4) with v given in Theorem 5.1, and assume $k_t(\eta) = \prod_{x\in\eta} f_t(x)$ belongs to the domain of the operator V. Then, k_t is a solution to the Vlasov hierarchy equation (4.8).

Note that the mesoscopic description of the C-S model (5.4) with the Vlasov operator defined in Theorem 5.1 is exactly the same as the kinetic Cucker-Smale equation (1.4).

VI. CONCLUSION

We introduce the analogs of the Kuramoto model and C-S model for infinite interacting particle systems and derive the mesoscopic equations governing those two models. We first set up the dynamics on the observable spaces for each model and then convert its dynamics to quasiobservables via the duality. Then, for each model, we use the Vlasov scaling technique in Ref. 10 to obtain the corresponding mesoscopic equation. The derived equations are identical to the well-known kinetic descriptions for the Kuramoto and C-S models derived by using standard BBGKY hierarchy techniques for finite systems. The BBGKY hierarchy technique for finite systems as well as the approach introduced for the Kuramoto model and C-S model for infinite interacting particle systems is valid for the integrable particle density functions. Typically, the case of integrable correlation functions yields that effectively our stochastic dynamics evolves through finite configurations only. To improve this situation for the case of bounded correlation functions (that includes the Poisson case), one could try to realize the

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derivation scheme of the kinetic equation just by considering (3.5) in its weak form, i.e., as (3.3) which corresponds to the weak form of the FP equation (3.1). In this case, solutions to (3.5), (3.8), and (3.10) should be understood in terms of distributions with the class of test functions being some subclass of $B_{bs}(\Gamma_0)$. The Vlasov scaling scheme has to be justified for such a case separately.

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APPENDIX: MEAN-FIELD LIMIT OF THE KURAMOTO MODEL AND CUCKER-SMALE MODEL THROUGH FORMAL BBGKY HIERARCHY

In this appendix, we revisit the BBGKY hierarchy approach which was first introduced in Ref. 15. Since the derivation of the Kuramoto-Sakaguchi equation (1.3) is exactly the same as for the kinetic C-S equation with less calculation, we only mention here the derivation of the kinetic C-S equation. Let $f^N = f^N(x_1, \ldots, x_N, v_1, \ldots, v_N, t)$ be the N-particle density function in the state space $\mathbb{R}^{2dN}$. Then, the Liouville equation governing the particle density function is given by

$$\partial_t f^N + \sum_{i=1}^N v_i \cdot \nabla_{x_i} f^N + \sum_{i=1}^N \nabla_{v_i} \cdot \left[\frac{1}{N} \sum_{j=1}^N \phi(x_i - x_j)(v_j - v_i) f^N \right] = 0. \tag{A1}$$

We note that the slightly modified evolution equation $\frac{\partial}{\partial t} F_t = LF_t$, with L defined in (5.1), is indeed a dual equation to (A1). The modification concerns the multiplication factor $\frac{1}{N}$ for the sum which includes $\nabla_{v_x} F$. We now take integration over the variables $x_2, v_2, \ldots, x_N, v_N$ to obtain the equation for the marginal distribution, which is exactly the one particle distribution function,

$$\partial_t f^N + v_1 \cdot \nabla_{x_1} f^N + \left(1 - \frac{1}{N}\right) \nabla_{v_1} \cdot \left[\int_{\mathbb{R}^{2d}} \phi(x_1 - x_2)(v_2 - v_1) g^N \, dx_2 dv_2 \right] = 0.$$

Here, we use the same notation $f^N = f^N(x_1, v_1, t)$ for the one particle distribution function, and we use $g^N = g^N(x_1, v_1, x_2, v_2, t)$ to denote the two particle distribution function,

$$g^N(x_1, v_1, x_2, v_2, t) := \int_{\mathbb{R}^{2(N-2)d}} f^N(x_1, v_1, \ldots, x_N, v_N, t) \, dx_3 dv_3 \cdots dx_N dv_N.$$

By letting $N \to \infty$, we arrive at the equation for the limiting one- and two-particle distribution functions f and g, respectively, which is given by

$$\partial_t f + v_1 \cdot \nabla_{x_1} f + \nabla_{v_1} \cdot \left[\int_{\mathbb{R}^{2d}} \phi(x_1 - x_2)(v_2 - v_1) g \, dx_2 dv_2 \right] = 0. \tag{A2}$$

To close the equation, we need the "molecular chaos assumption," which tells us that the two-particle density function is a product of one-particle density functions,

$$g(x_1, v_1, x_2, v_2, t) = f(x_1, v_1, t) f(x_2, v_2, t). \tag{A3}$$

Then, we finally substitute (A3) to (A2) to get the following McKean-Vlasov type mean-field limit equation for the C-S particle distribution function:

$$\partial_t f + v \cdot \nabla_x f + \nabla_v \cdot \left[\int_{\mathbb{R}^{2d}} \phi(x - x_*)(v_* - v) f(x, v, t) f(x_*, v_*, t) \, dx_* dv_* \right] = 0.$$

REFERENCES

[1] S. Albeverio, Y. Kondratiev, and M. Röckner, "Analysis and geometry on configuration spaces," J. Funct. Anal. **154**, 444–500 (1998).
[2] V. Belavkin, V. Maslov, and S. Tariverdiev, "The asymptotic dynamics of a system with a large number of particles described by Kolmogorov-Feller equations," Theor. Math. Phys. **49**(3), 1043–1049 (1981).
[3] W. Braun and K. Hepp, "The Vlasov dynamics and its fluctuations in the 1/N limit of interacting classical particles," Commun. Math. Phys. **56**(2), 101–113 (1977).
[4] Y.-P. Choi, S.-Y. Ha, and Z. Li, "Emergent dynamics of the Cucker-Smale flocking model and its variants," in *Active Particles, Volume 1: Advances in Theory, Models, and Applications*, edited by N. Bellomo, P. Degond, and E. Tadmor (Birkhauser-Springer, Cham, 2017), pp. 299–331.
[5] F. Cucker and S. Smale, "Emergent behavior in flocks," IEEE Trans. Autom. Control **52**, 852–862 (2007).
[6] R. L. Dobrushin, "Vlasov equations," Funct. Anal. Appl. **13**(2), 115–123 (1979).
[7] D. Finkelshtein, Y. Kondratiev, and M. J. Oliveira, "Markov evolutions and hierarchical equations in the continuum. I: One-component systems," J. Evol. Equations **9**(2), 197–233 (2009).

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[8]D. Finkelshtein, Y. Kondratiev, and O. Kutoviy, "Individual based model with competition in spatial ecology," SIAM J. Math. Anal. **41**(1), 297–317 (2009).
[9]D. Finkelshtein, Y. Kondratiev, and O. Kutoviy, "Semigroup approach to birth-and-death stochastic dynamics in continuum," J. Funct. Anal. **262**, 1274–1308 (2012).
[10]D. Finkelshtein, Y. Kondratiev, and O. Kutoviy, "Vlasov scaling for stochastic dynamics of continuous systems," J. Stat. Phys. **141**(1), 158–178 (2010).
[11]S.-Y. Ha, D. Ko, J. Park, and X. Zhang, "Collective synchronization of classical and quantum oscillators," EMS Surv. Math. Sci. **3**, 209–267 (2016).
[12]S.-Y. Ha, J. Kim, J. Park, and X. Zhang, "Uniform stability and mean-field limit for the augmented Kuramoto model," Netw. Heterog. Media **13**, 297–322 (2018).
[13]S.-Y. Ha, H. Kim, and S. Ryoo, "Emergence of phase-locked states for the Kuramoto model in a large coupling regime," Commun. Math. Sci. **14**, 1073–1091 (2016).
[14]S.-Y. Ha, J. Kim, and X. Zhang, "Uniform stability of the Cucker-Smale model and its application to the mean-field limit," Kinet. Relat. Models **11**, 1157–1181 (2018).
[15]S.-Y. Ha and E. Tadmor, "From particle to kinetic and hydrodynamic descriptions of flocking," Kinet. Relat. Models **1**, 415–435 (2008).
[16]S.-Y. Ha and J.-G. Liu, "A simple proof of the Cucker-Smale flocking dynamics and mean-field limit," Commun. Math. Sci. **7**, 297–325 (2009).
[17]Y. Kondratiev and T. Kuna, "Harmonic analysis on configuration spaces," Infinite Dimens. Anal., Quantum Probab. Relat. Top. **05**, 201 (2002).
[18]Y. Kondratiev, O. Kutoviy, and R. Minlos, "On non-equilibrium stochastic dynamics for interacting particle systems in continuum," J. Funct. Anal. **255**(1), 200–227 (2008).
[19]Y. Kondratiev, O. Kutoviy, and E. Zhizhina, "Nonequilibrium Glauber-type dynamics in continuum," J. Math. Phys. **47**(11), 113501 (2006).
[20]Y. Kondratiev, E. Lytvynov, and M. Röckner, "Equilibrium Kawasaki dynamics of continuous particle systems," Infinite Dimens. Anal., Quantum Probab. Relat. Top. **10**(2), 185–209 (2007).
[21]Y. Kondratiev and A. Skorokhod, "On contact processes in continuum," Infinite Dimens. Anal., Quantum Probab. Relat. Top. **09**, 187–198 (2006).
[22]V. V. Kozlov, "The generalized Vlasov kinetic equation," Russ. Math. Surv. **63**(4), 691–726 (2008).
[23]Y. Kuramoto, *Chemical Oscillations, Waves and Turbulence* (Springer-Verlag, Berlin, 1984).
[24]Y. Kuramoto, *International Symposium on Mathematical Problems in Theoretical Physics*, Lecture Notes in Theoretical Physics Vol. 30 (Springer-Verlag, Berlin, 1975), p. 420.
[25]M. J. Oliveira, "Configuration space analysis and Poissonian white noise analysis," Ph.D. thesis, University of Lisbon, 2012.
[26]C. S. Peskin, *Mathematical Aspects of Heart Physiology* (Courant Institute of Mathematical Sciences, 1975), pp. 268–278.
[27]H. Spohn, "Kinetic equations from Hamiltonian dynamics: Markovian limits," Rev. Mod. Phys. **52**, 569–615 (1980).
[28]T. Vicsek, A. Czirók, E. Ben-Jacob, I. Cohen, and O. Schochet, "Novel type of phase transition in a system of self-driven particles," Phys. Rev. Lett. **75**, 1226–1229 (1995).
[29]A. T. Winfree, "Biological rhythms and the behavior of populations of coupled oscillators," J. Theor. Biol. **16**, 15–42 (1967).

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